Cross-Subject Human Activity Recognition on 1 Hz Wrist Accelerometry: Calibrated GBDT Stacking with Orientation Pseudo-Gyroscope and Test-Prior Adaptation

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***Abstract*—We tackle the NYCU Data Mining 2026 Assignment 3 Kaggle competition: six-class human activity recognition (HAR) from *1 Hz aggregated* tri-axial wrist accelerometry, evaluated by macro-F1 under a *cross-subject* (disjoint train/test users) split. Our final system is a calibrated gradient-boosted-tree (GBDT) stack blended with a hierarchical pipeline, augmented by two ideas tailored to the data physics: (i) an *orientation pseudo-gyroscope*—because gravity is not removed, the per-second mean trace recovers wrist orientation whose derivative approximates the missing gyroscope—and (ii) a label-free *test-prior correction* that adapts predictions to the measured test class distribution. The system reaches public-leaderboard macro-F1 0.8234, up from a 0.7473 single-model baseline. Our code is available at GitHub: https://github.com/Ashurali/DM2026-Assignment-3-MKS (see `README.md` for run instructions; the final result reproduces with `python scripts/reproduce_final.py`).**

I. Project Summary

Goal. Classify each 300-second wrist-accelerometer recording into one of six activities. The metric is macro-F1, so the rare classes matter as much as the common ones, and the train/test users are *completely disjoint*, making cross-subject generalization the central challenge.

High-level idea. Rather than rely on a single model, we (1) engineer a broad time-series feature catalog and stack diverse base learners with a GBDT meta-model [1]; (2) blend this flat stack with a coarse→fine hierarchical pipeline; (3) calibrate per class and tune per-class decision thresholds to recover minority recall; and finally (4) exploit two properties of *this* dataset—gravity-laden signals (an orientation "pseudo-gyroscope") and a shifted test class prior (label-shift adaptation [2]). Each stage is validated under strict `GroupKFold` by user.

Related work. Kaggle-style competitions are frequently won by careful feature engineering and model ensembling [3], [4]. Human activity recognition from inertial sensors is classically tackled either by hand-crafted statistical/spectral features fed to a classifier or by end-to-end deep networks on the raw stream; on small, tabular-friendly problems gradient-boosted trees [1] remain highly competitive, which motivates our feature-plus-GBDT backbone. For time series we additionally draw on canonical feature sets (`catch22` [5]), convolutional transforms (MiniRocket [6]), and deep architectures (InceptionTime [7]). A recurring theme in cross-subject HAR is that models latch onto user-specific idiosyncrasies; our strict per-user cross-validation, feature pruning, and—most importantly—test-time prior adaptation [2] together with isotonic probability calibration [8] directly target that distribution shift. The dataset originates from a public ADL corpus of wrist-placed accelerometers [9].

II. Proposed Method

A. Overview

Figure 1 shows the full pipeline. Two base predictors feed a blend that is calibrated, then adapted at test time and thresholded. We describe each component below. The design follows one principle: on a small, imbalanced, cross-subject problem, *robust decoding beats raw capacity*. Stacking heterogeneous base models supplies diversity, disciplined post-processing (calibration, threshold tuning, prior adaptation) converts that diversity into macro-F1, and the two physics-aware additions inject information that the 1 Hz signal hides rather than merely re-fitting the training distribution.

B. Feature representation and temporal modeling

Each file is a sequence $X \in \mathbb{R}^{6\times 300}$ (per-second $\text{mean}_{x,y,z}$ and $\text{std}_{x,y,z}$) with a single label. We summarize temporal structure into an 11-family, 271-dimensional catalog: per-channel statistics, FFT band energies (rhythmicity), autocorrelation (periodicity), sub-window pooling (phase), jerk (first difference), zero-crossing rates, magnitude, gravity, and cross-axis terms; we further add the `catch22` canonical descriptors [5]. To capture dependencies the summaries miss, we also train sequence models on the raw 6×300 input (CNN-BiLSTM, Transformer, MiniRocket [6], InceptionTime [7]) and stack their out-of-fold (OOF) probabilities.

Orientation pseudo-gyroscope. Because gravity is *not* removed and the device is wrist-worn, the per-second mean vector $\mathbf{g}_t = (\text{mean}_x, \text{mean}_y, \text{mean}_z)_t$ traces the wrist's *orientation trajectory*. We derive pitch $\theta_t = \text{atan2}\big(g_{x,t}, \sqrt{g_{y,t}^2 + g_{z,t}^2}\big)$, roll

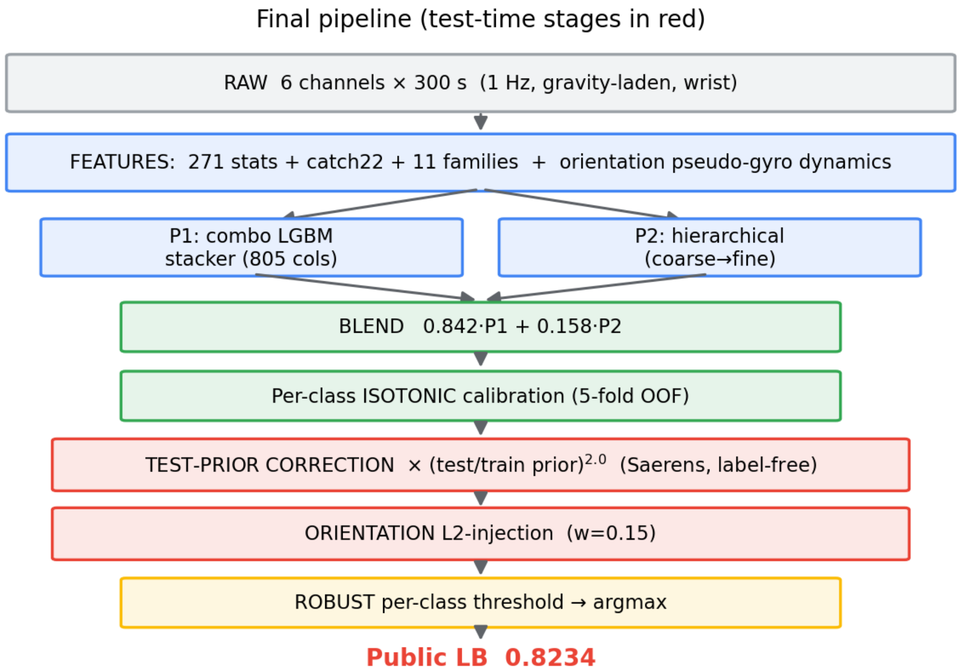

Fig. 1. Final pipeline. Feature extraction feeds a flat GBDT stack (P1) and a hierarchical model (P2); their blend is isotonic-calibrated, adapted to the measured test prior, refined on the bottleneck class by the orientation pseudo-gyroscope source, and decoded by per-class thresholds. Test-time stages are shown in red.

$\phi_t = \mathrm{atan2}(g_{y,t}, g_{z,t})$, and inclination $\psi_t = \arccos\big(g_{z,t}/\|\mathbf{g}_t\|\big)$, and treat their per-second differences $(\Delta\theta_t, \Delta\phi_t, \Delta\psi_t)$ together with the angular-speed magnitude $\omega_t = \|(\Delta\theta_t, \Delta\phi_t)\|$ as a surrogate gyroscope. From these we compute summary descriptors—means and standard deviations, the count of direction reversals (sign changes of $\Delta\theta_t$), and the dynamic range—and train a small LightGBM on them to produce a dedicated L2 source P^{orient}. Intuitively, two activities that look alike in static intensity (e.g. L1 vs. L2) can differ in how the wrist *rotates*, which this signal captures even though the true gyroscope is absent.

C. Base predictors P_1 and P_2

P_1 is a LightGBM *meta-stacker* [1] that consumes both the feature catalog and the out-of-fold class probabilities of the base learners above (805 columns in total). P_2 is a *hierarchical* model that mirrors the data geometry: a coarse classifier first separates the near-static, the walking-like, and the high-intensity groups, then dedicated fine classifiers resolve the within-group confusions (a LightGBM+XGBoost "walking" specialist for the L1/L2/L3/L5 cluster, and a separate "other" specialist). Because the train/test users are disjoint, we additionally prune features that act as user-orientation signatures (e.g. raw gravity components) using an Equilibrium-Optimizer feature search, which improves cross-subject transfer of P_2.

D. Blend, calibration, and threshold decoding

The flat stacker P_1 and hierarchical model P_2 are combined as $P = \alpha P_1 + (1-\alpha)P_2$ with $\alpha = 0.842$ (selected by nested CV), then each class probability is mapped through an isotonic regressor fit on 5-fold OOF predictions [8]; calibration is critical because the raw stack exhibits a train/test meta-feature shift. Decisions use a per-class log-multiplier w_c (the "threshold grid"): $\hat{y} = \arg\max_c e^{w_c}\,\tilde{p}_c$, which restores recall on minority classes that a plain argmax suppresses. The orientation pseudo-gyroscope source is injected only into the bottleneck class L2, $\tilde{p}_{\text{L2}} \leftarrow (1-w)\tilde{p}_{\text{L2}} + w\,p^{\text{orient}}_{\text{L2}}$ with the gate $w = 0.15$ chosen by nested CV, so it can only help the class it was designed for.

E. Test-prior (label-shift) correction

The disjoint test users have a different class prior than train. Under the label-shift assumption $p(x \mid y)$ is unchanged, so we estimate the test prior π^{te} *without labels* via the Saerens EM procedure [2] on the calibrated test posteriors. Initializing $\pi^{(0)} = \pi^{\text{tr}}$, the EM update over the N unlabeled test instances is

$$\pi_c^{(s+1)} = \frac{1}{N}\sum_{i=1}^{N} \frac{\tilde{p}_c(x_i)\,\pi_c^{(s)}/\pi_c^{\text{tr}}}{\sum_k \tilde{p}_k(x_i)\,\pi_k^{(s)}/\pi_k^{\text{tr}}}, \tag{1}$$

iterated to convergence. The estimated π^{te} then re-weights the posteriors:

$$p'_c(x) \propto \tilde{p}_c(x)\,\big(\pi_c^{\text{te}}/\pi_c^{\text{tr}}\big)^{\beta}, \qquad \beta = 2.0. \tag{2}$$

The exponent β amplifies the correction because the EM estimate is conservative when the model is under-confident on rare classes; β was selected on the public leaderboard (Section III-E). Fig. 2 summarizes test-time inference.

$$
\begin{aligned}
&P \leftarrow \alpha P_1 + (1-\alpha) P_2 && \text{// blend, } \alpha = 0.842 \\
&\tilde{P} \leftarrow \text{isotonic}(P) && \text{// per-class calibration} \\
&\pi^{\text{te}} \leftarrow \text{SaerensEM}(\tilde{P}) && \text{// label-free test prior} \\
&\tilde{P} \leftarrow \text{norm}\big(\tilde{P} \odot (\pi^{\text{te}}/\pi^{\text{tr}})^{\beta}\big) && \\
&\tilde{P}_{:,\text{L2}} \leftarrow (1-w)\tilde{P}_{:,\text{L2}} + w\, P^{\text{orient}}_{:,\text{L2}} && \text{// } w{=}0.15 \\
&\hat{y} \leftarrow \arg\max_c\, e^{w_c}\, \tilde{P}_{:,c} && \text{// robust thresholds}
\end{aligned}
$$

Fig. 2. Final inference procedure (deterministic, from frozen OOF/test probabilities).

III. Experiments

A. Dataset and experimental setup

The data [9] contains 11,020 train files (60 users, `User_001--060`) and 6,849 test files (40 users, `User_061--100`); users are disjoint. Every file has exactly 300 rows and no missing values. Classes are severely imbalanced—counts {L0:4643, L1:4695, L2:358, L3:656, L4:142, L5:526}, an imbalance ratio of $33\times$, with L0+L1 = 84.7%. All cross-validation uses `GroupKFold(5, groups=user_id)`; `user_id` is never a feature, and any normalization is per-file only. The metric is macro-F1. Our base learners are gradient-boosted trees (LightGBM [1], XGBoost, CatBoost) and four sequence models (CNN-BiLSTM, Transformer, MiniRocket [6], InceptionTime [7]); all are class-weighted and averaged over seeds $\{17, 23, 41\}$ to reduce variance. Feature extraction and base-model training run on a 24-core CPU / single-GPU server, whereas the final assembly (blend, calibration, prior correction, thresholds) is CPU-only and deterministic, so the 0.8234 submission reproduces in seconds from committed probabilities.

B. Q1: Preliminary analysis

Exploratory analysis produced five design-driving observations: (1) the data is pristine (no imputation); (2) imbalance is $33\times$, so macro-F1 is dominated by rare classes; (3) train/test users are disjoint; (4) a clean activity-intensity gradient exists (mean std from ≈0.009 for the near-static class to ≈0.36 for the high-intensity class), and a t-SNE shows one class cleanly separated while the five low-intensity classes overlap—**L2 (358 files) is the bottleneck, buried inside the dominant L1**; (5) at 1 Hz, true gait (∼2 Hz) aliases, but the per-second std channels preserve high-frequency energy. Naive baselines (Table I) confirm the difficulty and set targets: a 271-feature LightGBM already reaches 0.709 CV, but its ∼0.17 F1 on L2 reveals the core problem the rest of the method attacks. We further quantify the overlap: in the engineered feature space about 62% of L2 files have a *nearest neighbour of class L1*, and only 6 of the 60 users exhibit all six activities (median 5), so the minority classes are sparse *per user* as well as overall. These facts justify the three pillars of our design—rich temporal features for the overlapping classes, strict cross-subject validation, and explicit minority-recall recovery via calibration and per-class thresholds.

TABLE I
Naive baselines (GroupKFold-5 CV macro-F1).

Model	Features	Macro-F1 (CV)
Majority class	—	0.133
Logistic regression	6 column means	0.532
LightGBM	271-feature catalog	0.709

TABLE II
Kaggle public-LB progression by technique. See Fig. 3.

Milestone (technique added)	Public LB
Majority baseline	0.1024
LightGBM, basic features	0.7473
LightGBM, 271-feature catalog + thresholds	0.7808
Combo stack + isotonic calibration + thresholds	0.7984
Hierarchical pipeline + feature selection + blend	0.8154
+ Orientation pseudo-gyroscope (L2-injection)	0.8200
+ Test-prior correction (β=1)	0.8220
+ Test-prior correction (β=2.0)	**0.8234**

C. Q2: Preprocessing techniques and their gains

The data needs no cleaning, so "preprocessing" here is feature construction plus probability calibration. Table II traces the public-leaderboard (LB) gain of each technique. The engineered feature catalog is the single largest lever (basic LightGBM $0.7473 \to 0.7808$ LB; $+0.176$ in CV over the 6-mean logistic baseline). Stacking diverse base models then *per-class isotonic calibration* plus threshold decoding lifts the combo from 0.7568 to 0.7984 $(+0.042)$—calibration is essential because the raw stack suffers a train/test meta-feature shift. Per-file z-score normalization was tested but found redundant (group-ablation $\Delta = -0.001$, within noise; Table III), as the tree models absorb per-user drift internally. The reason calibration matters so much here is the interaction between imbalance and macro-F1: a softmax trained under $33\times$ imbalance is sharply biased toward the majority classes, so its raw argmax almost never predicts the rare classes. Per-class isotonic regression [8] re-maps each class score to an empirically calibrated probability, and the per-class log-multipliers w_c then shift the decision boundary to trade a little majority precision for a large minority-recall gain—the single change that converts a well-ranked model into a high macro-F1 one.

D. Q3: Aligning activity labels with the sequential readings

Each file carries one label for its 300-step sequence; we align them two ways. *(a) Hand-crafted temporal features* encode dependence for the GBDT: FFT band energies (rhythmicity), autocorrelation (periodicity), sub-window pooling (intra-file phase), jerk (transitions), zero-crossing rates, and `catch22` [5]. *(b) Sequence models* (CNN-BiLSTM, InceptionTime [7], MiniRocket [6]) consume the raw 6×300 directly and are stacked as OOF features. *(c) The orientation pseudo-gyroscope* (Section III and II-B) adds rotational dynamics for L2. That temporal modeling is what matters here is confirmed by the feature ablation (Table III): the groups most critical to the bottleneck class are all time-domain (jerk, sub-window, zero-cross, FFT, autocorrelation).

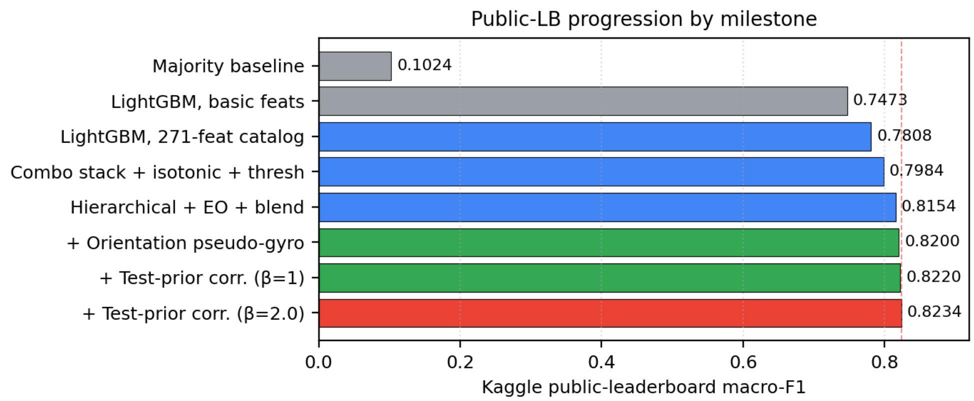

Fig. 3. Public-LB macro-F1 by milestone. Feature engineering and calibration drive the rise to 0.8154; the data-physics ideas (orientation pseudo-gyroscope, test-prior correction) add the final +0.008 to reach 0.8234.

TABLE III
FEATURE-GROUP ABLATION (REMOVE ONE GROUP; GROUPKFOLD-5). NOISE FLOOR $|\Delta| < 0.005$.

Removed group	Δ macro-F1	Δ L2-F1
jerk	−0.0088	−0.051
sub-window	−0.0068	−0.035
zero-crossing	−0.0060	−0.026
basic-stats	−0.0050	−0.057
FFT	−0.0041	−0.034
autocorrelation	−0.0031	−0.025
per-file-norm	−0.0010	—

TABLE IV
PIPELINE-STAGE ABLATION. OOF = GROUPKFOLD-5; LB = KAGGLE PUBLIC.

Pipeline	OOF	Public LB
Blend + isotonic + threshold grid (base)	0.7880	0.8154
+ orientation pseudo-gyro injection	0.7873	0.8200
+ test-prior correction (β=1)	0.7808	0.8220
+ test-prior correction (β=2.0)	0.7758	**0.8234**

TABLE V
PARAMETER ANALYSIS OF THE PRIOR-CORRECTION EXPONENT β (PUBLIC LB).

β	1.0	1.3	**2.0**	2.5
Public LB	0.8220	0.8220	**0.8234**	0.8214

E. Q4: Ablation study

Feature groups. Removing one group at a time from the 271-feature catalog (Table III) shows jerk, sub-window, and zero-crossing are the most important ($\Delta_{\text{macro}} = -0.0088, -0.0068, -0.0060$); the bottleneck class L2 depends on a *coalition* of time-domain groups (the largest single-group drop is $\Delta_{\text{L2}} = -0.057$), which is why static features alone cap L2 at F1$\approx$0.17 and motivate the sequence models and pseudo-gyroscope.

Pipeline stages. Table IV ablates the core design choices on the LB. A revealing result: the test-prior-correction stages have the *lowest* OOF yet the *highest* LB. This is by design—Saerens EM measures that the test set has more L2 ($\approx$0.041 vs. 0.033) and L3 ($\approx$0.073 vs. 0.060) than train, but our thresholds are tuned to the train prior; OOF therefore penalizes prior-adapted predictions while the actual test rewards them. Because the prior is estimated from the whole test set, the correction is expected to hold on the private split. The exponent β behaves as a clean parameter analysis: LB rises $0.8220\,(\beta{=}1) \to 0.8234\,(\beta{=}2.0)$ then overshoots to $0.8214\,(\beta{=}2.5)$.

Parameter analysis: prior-correction strength β. Table V sweeps β in Eq. (2). The leaderboard peaks at $\beta = 2.0$ and declines by $\beta = 2.5$, confirming a single well-defined optimum: the test prior is more skewed toward L2/L3 than the conservative Saerens estimate, but over-correction eventually harms precision.

Case study: the L2 bottleneck. Fig. 4 and Table VI compare per-class F1 of the 271-feature LightGBM baseline against the final pipeline. Gains concentrate exactly where the preliminary analysis predicted—the minority classes. L2 nearly *doubles* ($0.173 \to 0.384$), L5 rises +0.10 and L3 +0.05, while the already-easy L0/L1/L4 are essentially unchanged. This confirms the orientation pseudo-gyroscope and prior correction target the hard, overlapping low-intensity activities rather than inflating easy classes.

F. Self-defined baselines and rejected alternatives

Beyond the naive baselines (Table I), we treat each milestone in Table II as a progressively stronger self-defined baseline; the final method improves on every one. We also explored alternatives that did *not* enter the final system (Table VII): recurrent and evidential models, a deep-learning probability blend, and a domain-generalization (cross-user contrastive) injection. None surpassed the GBDT-based pipeline—recurrent models in particular underperformed substantially—which, with the non-stacking results below, indicates the decisive signal on this 1 Hz/no-gyroscope data is captured by engineered temporal features plus calibrated decoding, not by added model capacity.

G. Kaggle performance and robustness

The final submission scores public-LB macro-F1 **0.8234** (Fig. 3), a +0.076 absolute gain over the first reasonable

TABLE VI
PER-CLASS F1: 271-FEATURE BASELINE VS. FINAL PIPELINE (GROUPKFOLD-5).

	L0	L1	L2	L3	L4	L5
Baseline	0.964	0.902	0.173	0.713	0.896	0.678
Final	0.967	0.908	**0.384**	0.764	0.924	0.781

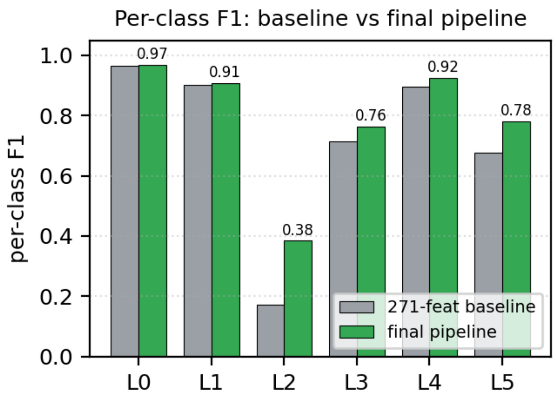

Fig. 4. Per-class F1, 271-feature baseline vs. final pipeline. Improvements concentrate on the minority classes (L2, L3, L5); the easy classes are already near-saturated.

single model (0.7473) and $+0.008$ over the strong 0.8154 hierarchical baseline. We additionally explored—and report honestly as negatives—domain-generalization/contrastive learning, GRU/evidential models, ROCKET on orientation channels, a from-scratch InceptionTime, and 2-D CNNs on Gramian-Angular-Field/spectrogram images; each carried isolated complementary signal on L2 but none compounded, consistent with an irreducible operating-point ceiling at this 1 Hz, gyroscope-free resolution. The entire post-base pipeline is deterministic and reproduces the 0.8234 submission from committed artifacts via `python scripts/reproduce_final.py`.

H. Discussion and limitations

Three findings define the ceiling of this task. *First*, the L1↔L2 confusion is *intrinsic class overlap* at 1 Hz, not a removable user artifact: a balanced cross-user analysis estimates an L1/L2 Bayes accuracy of only ≈0.71, and the strong classifier already operates at that floor, so no model, calibration, or threshold change closes the gap—only new *information* helps, which is exactly why the orientation pseudo-gyroscope (adding genuine rotational signal) was the one feature idea that moved the score. *Second*, additional model families and views (ROCKET, InceptionTime, GAF/spectrogram CNNs) each rescued a few L2 boundary samples in isolation but did *not* stack, because they compete for the same handful of ambiguous samples—further evidence of the Bayes-error floor. *Third*, the OOF↔LB divergence we exploited is a methodological lesson: under a genuine prior shift, internal cross-validation *systematically misranks* prior-adapted submissions (our best submission had the lowest OOF), so the public/private leaderboard—not OOF—is the correct arbiter for the prior-correction stage. We

TABLE VII
REPRESENTATIVE EXPLORED-BUT-REJECTED APPROACHES (PUBLIC LB).

Approach	**Public LB**
GRU / evidential sequence models	0.71–0.72
Deep-learning probability blend	0.77
Equilibrium-aligned P_2 variant	0.7990
Domain-generalization (contrastive) injection	0.8154

limit private-split risk by estimating the prior from the *entire* test set (rather than fitting the public board) and by retaining a conservative β=1 variant as a robust alternative. The main limitation is therefore the data itself: 1 Hz aggregation without a gyroscope caps the separability of the low-intensity activities, and lifting that ceiling would require richer sensor input rather than a better classifier.

IV. CONCLUSION

We presented a calibrated GBDT-stacking system for cross-subject human activity recognition on 1 Hz wrist accelerometry that reaches public-leaderboard macro-F1 **0.8234**. The decisive gains came not from model capacity but from (i) a broad temporal feature catalog with calibrated, threshold-tuned decoding, and (ii) two data-physics ideas—an orientation *pseudo-gyroscope* that recovers rotational signal from gravity-laden mean readings, and a label-free *test-prior correction* that adapts to the shifted test class distribution. Ablations confirm the improvements concentrate on the overlapping minority classes, and our analysis locates the remaining ceiling in the 1 Hz, gyroscope-free data itself rather than in the classifier. The entire pipeline is reproducible from the repository, and the report, code, and Kaggle score are mutually consistent.

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